The Hessian and Multivariate Taylor Series

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Eigenvectors and eigenvalues; Partial derivatives and gradients.
- **Leads to:** Convex and non-convex loss; XGBoost; iLQR and DDP.

1. Overview

Key point: The Hessian collects all second partial derivatives and measures how a surface curves; together with the gradient it builds the **multivariate Taylor polynomial** (G-1284), which approximates a function near a point by a plane, then a curved bowl, and so on.

This Note follows Chapter 5 (Sections 5.7 and 5.8) of *Mathematics for Machine Learning* (Deisenroth, Faisal and Ong, 2020).

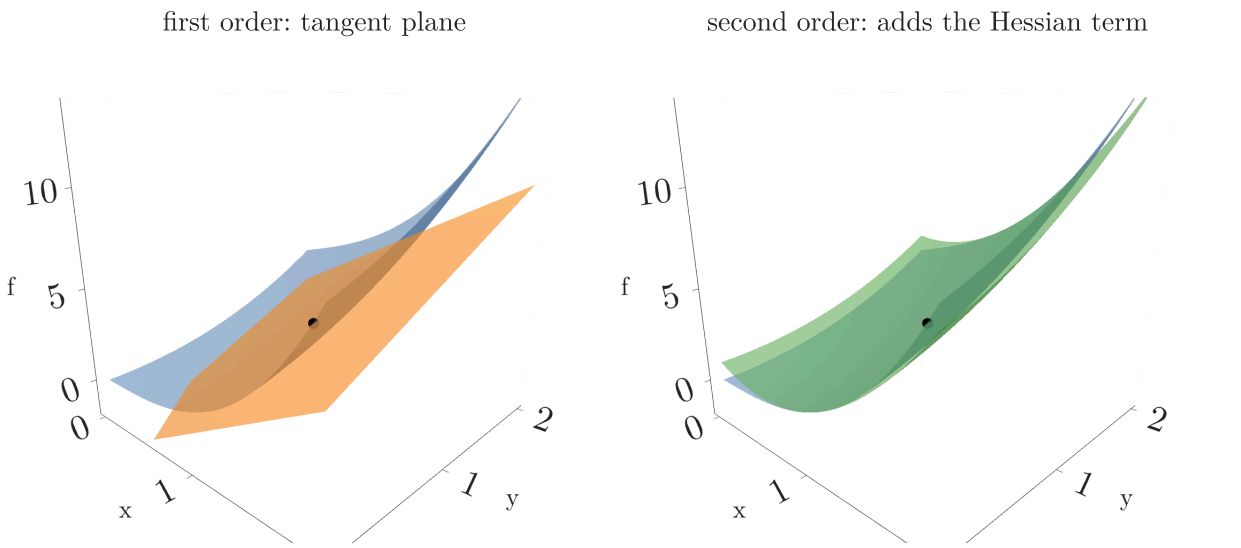


Figure 1: The function $f(x, y) = x^3 + xy + y^2$ (blue) near the point $(1, 1)$, approximated by its tangent plane (left, orange) and by its second-order Taylor polynomial (right, green)

Figure 1 shows the goal. Near a point, a surface is approximated first by a flat **tangent plane** (G-1946; left), which only uses the gradient. Adding a term built from second derivatives, the Hessian, bends the approximation so that it follows the surface over a much wider area (right).

The Taylor series in XGBoost did this with one variable: it called the second derivative of one **observation**'s (G-1374) loss (an observation is one record, a row of the data table) its Hessian h_i and stopped the Taylor series at the second-order term. This Note does the same with many variables, in this order:

1. how one curve bends: the second derivative (Section 2);
2. the flat approximation of a surface: the tangent plane (Section 3);
3. the curved approximation, which needs **second partial derivatives** (G-1760) (Section 4);
4. the Hessian, the matrix that holds those second partial derivatives and tells the shape of the surface (Section 5);
5. the full Taylor series and its uses in ML (Sections 6 and 7).

The Note uses the gradient (the list of partial derivatives, the slopes along each input) and eigenvalues (the numbers by which a matrix stretches its special directions).

2. The second derivative: how a curve bends

> **Key point:** The second derivative is the rate of change of the slope: positive where a curve bends upward, negative where it bends downward, and larger where the bend is sharper.

The derivative of a curve is its slope. The slope is itself a function, so it has a derivative too: the **second derivative**, written $f''(x)$ or $\dfrac{d^2f}{dx^2}$. Where a curve bends upward its slope is increasing, so $f'' > 0$. Where it bends downward its slope is decreasing, so $f'' < 0$. A sharper bend means a larger f''.

Figure 2 reads the second derivative off two steps. Take two equal steps dx to the right and compare the two rises.

1. The first step raises f by df_1, the second by df_2.
2. The difference $df_2 - df_1$ is the change in the change. On a straight line it is 0; on a curve that bends upward the second rise is bigger.
3. The second derivative is that difference divided by dx^2, in the limit of tiny steps.

For $f(x) = x^3$ from $x = 1$ with $dx = 0.01$, one step per line:

$$f(1) = 1$$

$$f(1.01) = 1.030301$$

$$f(1.02) = 1.061208$$

$$df_1 = 1.030301 - 1 = 0.030301$$

$$df_2 = 1.061208 - 1.030301 = 0.030907$$

$$df_2 - df_1 = 0.000606$$

$$\frac{0.000606}{0.01^2} = \frac{0.000606}{0.0001} = 6.06$$

This is close to the exact second derivative, $f''(x) = 6x$, which is 6 at $x = 1$.

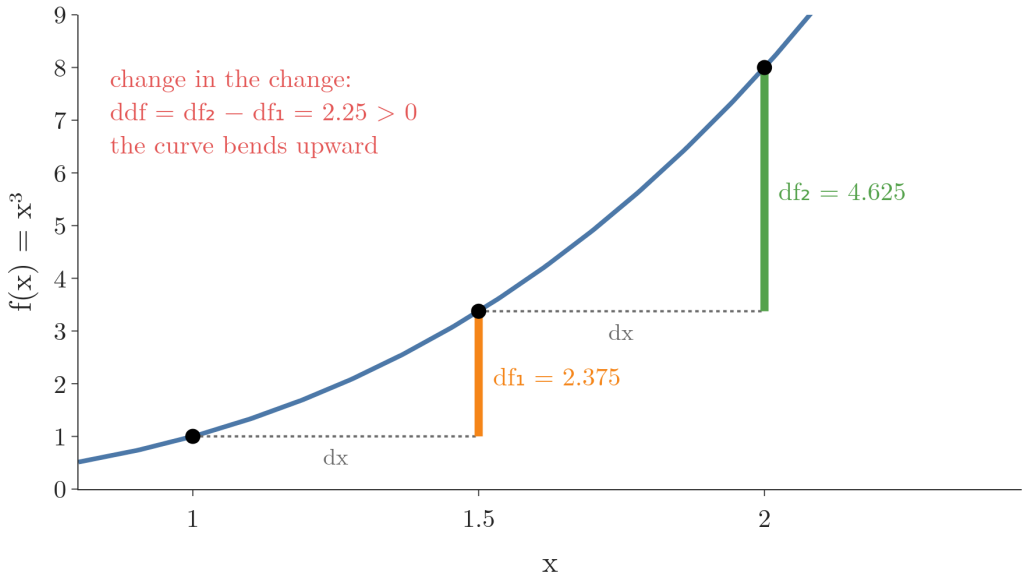


Figure 2: The second derivative as the change in the change, for $f(x) = x^3$ from $x = 1$. Two equal steps (drawn large, $dx = 0.5$) raise f by 2.375 and then by 4.625; the second rise is bigger, so the curve bends upward. Idea after 3Blue1Brown, "Higher order derivatives"

An everyday reading: if f is the distance a car has covered, f' is its velocity and f'' is its acceleration. A positive second derivative is the push back into the seat while the car speeds up; a negative one is braking.

Building a Taylor polynomial by hand used f'' to make a parabola bend like a curve. The rest of this Note does the same for a surface.

3. Linearisation: the tangent plane

> **Key point:** Near a point, a surface is close to the flat plane that touches it there. The plane's height is the value at the point plus the gradient times the step.

The tangent line replaced a curve near a point by a straight line. With several inputs, the gradient gives a tangent plane (Figure 1, left): the plane spanned by the tangent lines of the two slices.

Our running example is

$$f(x, y) = x^3 + xy + y^2$$

We approximate it near the point $(1, 1)$. In bold, the starting point is $\mathbf{x}_0 = (1, 1)$, the point we want is $\mathbf{x}$, and the step between them is $\mathbf{x} - \mathbf{x}_0$. For example, to reach $\mathbf{x} = (1.1, 0.9)$ the step is $(0.1, -0.1)$.

1. **Example first,** one step per line. The value at the point:

   $$f(1, 1) = 1 + 1 + 1 = 3$$

   The partial derivatives are

   $$\frac{\partial f}{\partial x} = 3x^2 + y$$
   $$\frac{\partial f}{\partial y} = x + 2y$$

   so at $(1, 1)$:

   $$\frac{\partial f}{\partial x} = 3 + 1 = 4$$
   $$\frac{\partial f}{\partial y} = 1 + 2 = 3$$
   $$\nabla f(1, 1) = [4,\ 3]$$

   The step $(0.1, -0.1)$ changes the height by one product per input:

   $$x \text{ part:} \quad 4 \times 0.1 = 0.4$$

   $$y \text{ part:} \quad 3 \times (-0.1) = -0.3$$
   $$f(1.1, 0.9) \approx 3 + 0.4 - 0.3 = 3.1$$

   The true value is 3.131.
2. **In words:** start from the value at $\mathbf{x}_0$ and add the gradient times the step. The gradient is a row and the step a column, so their product is one number: the sum of the products above.
3. **Formula:**

   $$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \nabla_{\mathbf{x}} f(\mathbf{x}_0)\,(\mathbf{x} - \mathbf{x}_0)$$

   Check with the numbers of step 1:

   $$3 + [4,\ 3] \cdot (0.1,\ -0.1)$$

   $$= 3 + 0.4 - 0.3 = 3.1$$

The step $\mathbf{x} - \mathbf{x}_0$ is what makes the plane touch the surface: at $\mathbf{x} = \mathbf{x}_0$ the step is zero, the second term vanishes, and the approximation equals $f(\mathbf{x}_0)$ exactly. The formula has the same form for 2 inputs or for 100.

The approximation is good close to $\mathbf{x}_0$ and gets worse further away, because a plane cannot bend. In Figure 1 (left), the orange plane touches the blue surface at the black dot and separates from it in every direction.

Figure 3 measures how far. The error at a point is the vertical gap between the blue surface and the approximation in Figure 1. The figure builds the picture in two rows:

- **Top row: the error as a surface.** Above each point (x, y) the height is the size of the error. For the tangent plane (left) the error is a bowl: 0 at $(1, 1)$, rising quickly in every direction. For the second-order polynomial (right) it is an almost flat trough.
- **Bottom row: the same surfaces seen from straight above,** as a heat map: the colour at each point shows the height, darker red = larger error. A heat map is a contour map (G-468: the surface seen from above) with colour in place of lines. The black line joins the points where the error is exactly 0.05, as the thin black lines on the surfaces above do: a closed oval for the tangent plane (left) and two straight vertical lines for the second-order polynomial (right). The colour bar on the right gives the error, from 0 (white) to 1 (dark red).

For the tangent plane (left), the error stays below 0.05 only inside a small oval around the point. The second-order polynomial of Section 4 (right) stays below 0.05 over a band about nine times as large (inside the plotted square; the band runs on up and down beyond it). Its only missing term is δ_x^3 (Section 6.3), so its error depends on x alone, which is why the band has straight sides.

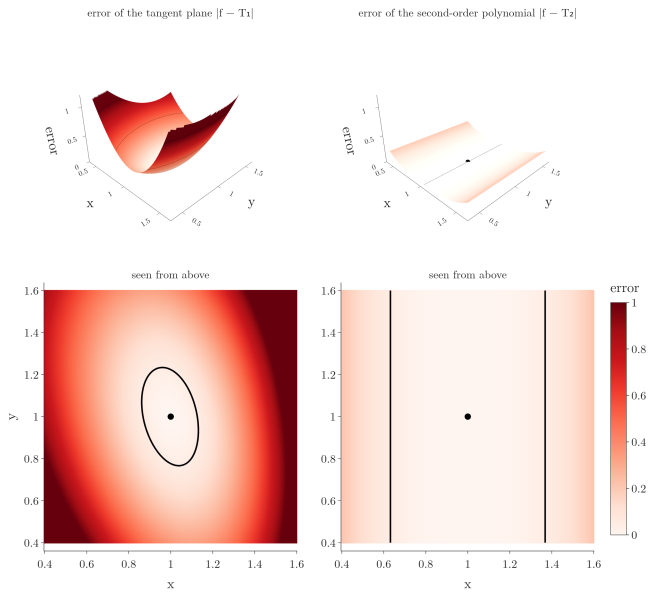


Figure 3: The size of the error near (1, 1), on the same colour scale. Top: each error as a surface (height = error). Bottom: the same seen from above as a heat map. Left: the tangent plane, $|f - T_1|$. Right: the second-order Taylor polynomial, $|f - T_2|$. The black line marks an error of 0.05; the region inside it is about nine times larger for T_2.

4. Bending the plane: the quadratic approximation

> **Key point:** Add three bending terms to the tangent plane (the step in x squared, the two steps multiplied, the step in y squared), and choose their constants so that the surface bends like f at the point.

A plane cannot bend, so the tangent plane leaves the surface quickly. The fix is a surface that can bend: the green surface of Figure 1 (right). It touches f at the point like the plane, and it also curves with f, so it stays close over a much wider area. Cut it with any vertical plane and the cut is a parabola.

Write the step as $\delta_x = x - x_0$ and $\delta_y = y - y_0$. The tangent plane uses the step once (δ_x, δ_y). The simplest terms that bend are the ones that use it twice:

$$Q(x, y) = \underbrace{f(\mathbf{x}_0) + f_x\,\delta_x + f_y\,\delta_y}_{\text{tangent plane}} + a\,\delta_x^2 + b\,\delta_x\delta_y + c\,\delta_y^2$$

Here f_x and f_y are short for the two partial derivatives at the point. Such a Q is a **quadratic approximation** (G-2252). The three new terms are zero at the point, and so are their first partial derivatives, so they do not disturb what the tangent plane already matches: the value and the gradient. The constants a, b and c are free. To choose them we need second derivatives of a function of two variables.

4.1 Second partial derivatives

> **Key point:** Differentiating a partial derivative again gives a second partial derivative; for smooth functions, the order of the two differentiations does not matter.

A partial derivative is itself a function of all the inputs, so we can differentiate it again. For a function $f(x, y)$ of two variables the notation is:

- $\partial^2 f/\partial x^2$: differentiate with respect to x twice;
- $\partial^2 f/\partial y\,\partial x$, the derivative with respect to y of $\partial f/\partial x$: first with respect to x, then with respect to y (read right to left);
- $\partial^n f/\partial x^n$: differentiate n times with respect to x.

Each of these is a **second partial derivative** (G-1760). The ones that mix two different variables are **mixed partial derivatives** (G-1236).

1. **In words:** for a function whose second derivatives are continuous, differentiating by x then y gives the same as by y then x.
2. **Formula:**
$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x}$$

3. **Example:** $f(x, y) = x^3 + xy + y^2$, our running example. Its first partial derivatives are $\partial f/\partial x = 3x^2 + y$ and $\partial f/\partial y = x + 2y$. Then:

$$\frac{\partial^2 f}{\partial x^2} = 6x$$

$$\frac{\partial^2 f}{\partial y^2} = 2$$

$$\frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial}{\partial y}(3x^2 + y) = 1$$

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial}{\partial x}(x + 2y) = 1$$

The two mixed derivatives agree.

Figure 4 shows what "agree" means. On the left, we stand at $x = 1$ and watch the x-slope $\partial f/\partial x$ as y moves: it rises 1 for every unit of y. On the right, we stand at $y = 1$ and watch the y-slope as x moves: it also rises 1 per unit. How the x-slope responds to y equals how the y-slope responds to x.

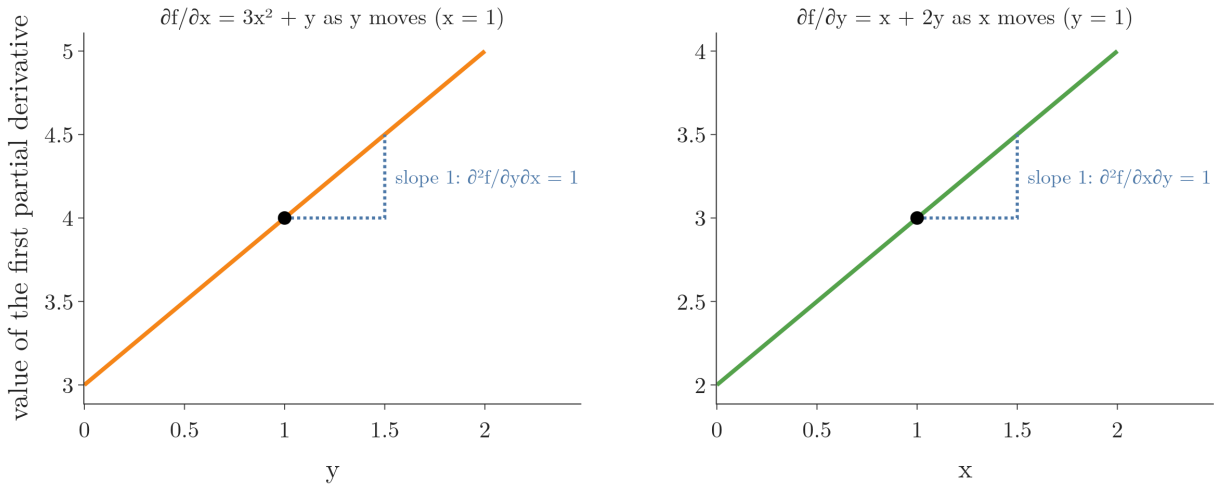


Figure 4: Mixed partial derivatives of $f(x, y) = x^3 + xy + y^2$ near $(1, 1)$. Left: the x-slope $3x^2 + y$ as y moves, with $x = 1$ fixed: slope 1. Right: the y-slope $x + 2y$ as x moves, with $y = 1$ fixed: also slope 1.

4.2 Fitting the three constants

> **Key point:** Matching the **second partial derivatives** (G-1760) gives $a = \frac{1}{2}f_{xx}$, $b = f_{xy}$ and $c = \frac{1}{2}f_{yy}$.

We want Q to bend like f at the point, so we ask for the same three second partial derivatives. For the running example at $(1, 1)$: $f = 3$, $\nabla f = [4, 3]$, and from Section 4.1 the second partial derivatives are 6, 1 and 2. So

$$Q = 3 + 4\delta_x + 3\delta_y$$
$$+a\,\delta_x^2 + b\,\delta_x\delta_y + c\,\delta_y^2$$

1. **Twice by x.** Only $a\,\delta_x^2$ survives two derivatives by x, and it gives $2a$. Matching $\partial^2 f/\partial x^2 = 6$ gives $a = 3$.
2. **By x, then by y.** Only $b\,\delta_x\delta_y$ survives, and it gives b. Matching the **mixed partial derivative** (G-1236) 1 gives $b = 1$.
3. **Twice by y.** Only $c\,\delta_y^2$ survives, and it gives $2c$. Matching $\partial^2 f/\partial y^2 = 2$ gives $c = 1$.

The result is

$$Q = 3 + 4\delta_x + 3\delta_y$$
$$+3\delta_x^2 + \delta_x\delta_y + \delta_y^2$$

Check at $(1.1, 0.9)$, a step of $\delta_x = 0.1$, $\delta_y = -0.1$. The plane part gave 3.1. The new terms, one per line:

$$3\delta_x^2 = 3 \times 0.01 = 0.03$$

$$\delta_x\delta_y = 0.1 \times (-0.1) = -0.01$$

$$\delta_y^2 = (-0.1)^2 = 0.01$$

$$0.03 - 0.01 + 0.01 = 0.03$$

$$Q = 3.1 + 0.03 = 3.13$$

against the true 3.131.

In general, with f_{xx}, f_{xy} and f_{yy} short for the three second partial derivatives at the point,

$$a = \tfrac{1}{2} f_{xx}$$
$$b = f_{xy}$$
$$c = \tfrac{1}{2} f_{yy}$$

The halves cancel the 2 that the power rule brings down from the squares, exactly as for the parabola of one variable. The mixed term has no square, so it has no half.

Three numbers control the bend of a function of two inputs. With n inputs there are many more, and they need a tidy home: the Hessian.

5. The Hessian

> **Key point:** The Hessian is the square, symmetric matrix of all **second partial derivatives** (G-1760); it describes the **curvature** (G-521) of f near a point.

5.1 The Hessian matrix

> **Key point:** Entry (i, j) is $\partial^2 f / \partial x_i \partial x_j$; for n inputs the Hessian is $n \times n$.

Figure 5 lays out every **second partial derivative** (G-1760) of the running example as a tree. Each arrow differentiates once, by x or by y. Two arrows in a row give four results, and the two mixed paths in the middle meet in the same value, as Section 4.1 showed. The four results, written in a grid, are the Hessian.

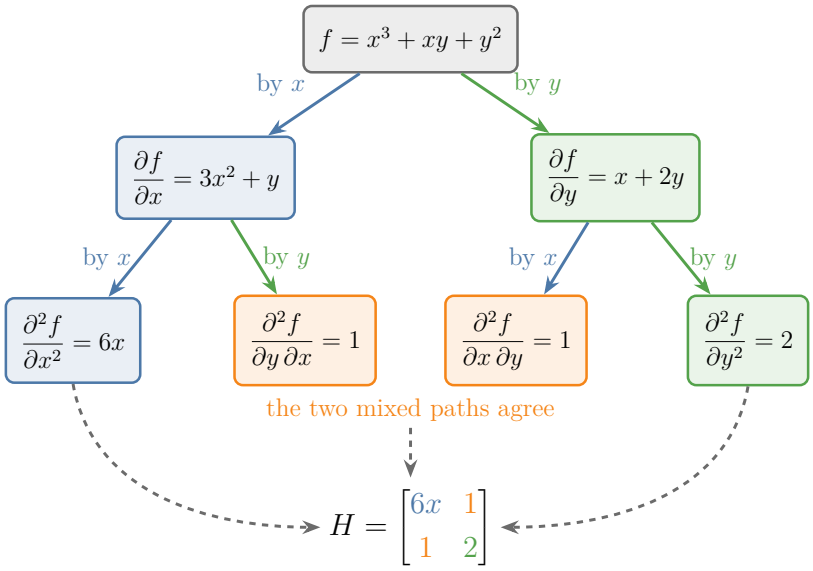


Figure 5: The tree of derivatives of $f = x^3 + xy + y^2$. Two differentiations give four second partial derivatives; the two mixed paths agree. The four results fill the Hessian. Idea after Khan Academy, "Symmetry of second partial derivatives" and "The Hessian matrix"; our own function

1. **In words:** the **Hessian matrix** (G-888) arranges all second partial derivatives in a grid, row i and column j holding the derivative with respect to x_i and x_j.
2. **Formula:** for two variables,

$$H = \nabla^2 f = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x^2} & \dfrac{\partial^2 f}{\partial x\,\partial y} \\ \dfrac{\partial^2 f}{\partial y\,\partial x} & \dfrac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

   For $f : \mathbb{R}^n \to \mathbb{R}$ it is an $n \times n$ matrix, symmetric because the mixed derivatives are equal.

3. **Example:** for $f(x, y) = x^3 + xy + y^2$,

$$H = \begin{bmatrix} 6x & 1 \\ 1 & 2 \end{bmatrix}$$

At the point $(1, 1)$:

$$H(1, 1) = \begin{bmatrix} 6 & 1 \\ 1 & 2 \end{bmatrix}$$

The Hessian is a matrix-valued function: put in a point, get a matrix of numbers. It also writes the three bending terms of Section 4.2 as one product. With the step as a column $\boldsymbol{\delta} = [\delta_x, \delta_y]^\mathsf{T}$,

$$\tfrac{1}{2}\,\boldsymbol{\delta}^\mathsf{T} H \boldsymbol{\delta} = \tfrac{1}{2}\left(6\delta_x^2 + 2\delta_x\delta_y + 2\delta_y^2\right)$$

$$= 3\delta_x^2 + \delta_x\delta_y + \delta_y^2$$

the same $a = 3$, $b = 1$, $c = 1$. The mixed entry sits in the matrix twice, which is why b needed no half. Section 6.1 uses this form.

The Hessian is also the Jacobian of the gradient: the gradient is a function with n outputs, and differentiating each of them by each input gives $n \times n$ numbers (see the Jacobian). With a single input, the Hessian is just the second derivative, the h_i of the second-order approximation of the objective.

> **Extra:** For a vector-valued function $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^m$, each of the m outputs has its own $n \times n$ Hessian. Stacked, they form an $m \times n \times n$ tensor.

5.2 Two Hessians we already know

> **Key point:** A bowl made only of squares and products of the inputs bends the same amount everywhere, so its Hessian is the same at every point. Two such bowls are the bowl of the contour-map section and the least-squares loss.

In plain words: the second derivative of x^2 is 2 at every x. A parabola bends by the same amount everywhere. Functions built only from squares and products of two inputs (quadratic functions) behave the same way: their Hessian is a fixed matrix of numbers.

- **The bowl** of the gradient on the map,

$$f(x_1, x_2) = x_1^2 + x_1x_2 + 2x_2^2$$

$$f(1, 1) = 4$$

has gradient $[2x_1 + x_2,\ x_1 + 4x_2]$. Differentiating each entry again, one per line:

$$\frac{\partial}{\partial x_1}(2x_1 + x_2) = 2$$

$$\frac{\partial}{\partial x_2}(2x_1 + x_2) = 1$$
$$\frac{\partial}{\partial x_1}(x_1 + 4x_2) = 1$$
$$\frac{\partial}{\partial x_2}(x_1 + 4x_2) = 4$$

No x_1 or x_2 is left, so the Hessian is the same at every point:

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$$

Its surface and contour map (in that section) show nested ellipses that all have the same shape: the same bend everywhere.

- **The least-squares loss** $\|\mathbf{y} - \Phi\boldsymbol{\theta}\|^2$ has gradient $2\boldsymbol{\theta}^\mathsf{T}\Phi^\mathsf{T}\Phi - 2\mathbf{y}^\mathsf{T}\Phi$ (the gradient of the least-squares loss). Its Jacobian is

$$H = 2\Phi^\mathsf{T}\Phi$$

For the three-point data there, Φ has rows $[1, 1]$, $[1, 2]$, $[1, 3]$ (a 1 for the intercept, then the input). Each entry of $\Phi^\mathsf{T}\Phi$ multiplies two columns of Φ entry by entry and adds:

$$\text{ones with ones:} \quad 1 \times 1 + 1 \times 1 + 1 \times 1 = 3$$
$$\text{ones with inputs:} \quad 1 \times 1 + 1 \times 2 + 1 \times 3 = 6$$
$$\text{inputs with inputs:} \quad 1 \times 1 + 2 \times 2 + 3 \times 3 = 14$$
$$\Phi^\mathsf{T}\Phi = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$
$$H = 2\Phi^\mathsf{T}\Phi = \begin{bmatrix} 6 & 12 \\ 12 & 28 \end{bmatrix}$$

The matrix $X^\mathsf{T}X$ of the normal equation is, up to the factor 2, the Hessian of the loss.

5.3 Curvature: what the Hessian says about the shape

> **Key point:** The signs of the Hessian's eigenvalues give the shape near a flat point: all positive is a bowl (minimum), all negative a cap (maximum), mixed signs a saddle.

Why the shape matters. Training a model means finding the lowest point of a loss. At the lowest point the tangent plane is flat: the gradient is zero. A point with zero gradient is a **stationary point** (G-1879). A flat tangent plane is not enough, though. The top of a hill is flat too, and so is the middle of a horse's saddle. The second derivatives tell the three apart.

Slices. Cut the surface with a vertical plane through the point. The cut is a curve of one variable, and its second derivative says whether that slice bends up or down. Take $f = x^2 - y^2$ at the origin, where the gradient $[2x, -2y]$ is zero.

- The slice along x (keep $y = 0$) is x^2: it bends up, so from this side the point looks like a minimum.
- The slice along y (keep $x = 0$) is $-y^2$: it bends down, so from this side the point looks like a maximum.

The two directions disagree. Such a point is a **saddle point** (G-1718): neither a minimum nor a maximum. A curve of one variable cannot do this, because it has only one direction. Figure 6 turns the slicing plane through every direction. Walk a distance s from the origin in the direction at angle θ to the x-axis; the point reached is $x = s\cos\theta$, $y = s\sin\theta$. Along this slice, one step per line:

$$f = x^2 - y^2 = s^2\cos^2\theta - s^2\sin^2\theta$$

$$= s^2\left(\cos^2\theta - \sin^2\theta\right)$$

$$= \cos(2\theta)\, s^2$$

The last line uses the identity $\cos^2\theta - \sin^2\theta = \cos 2\theta$. The second derivative of the slice with respect to s is therefore $2\cos 2\theta$:

$$\theta = 0 \text{ (along } x\text{)}: \quad 2\cos 0 = 2$$

$$\theta = 45^\circ \text{ (diagonal)}: \quad 2\cos 90^\circ = 0$$

$$\theta = 90^\circ \text{ (along } y\text{)}: \quad 2\cos 180^\circ = -2$$

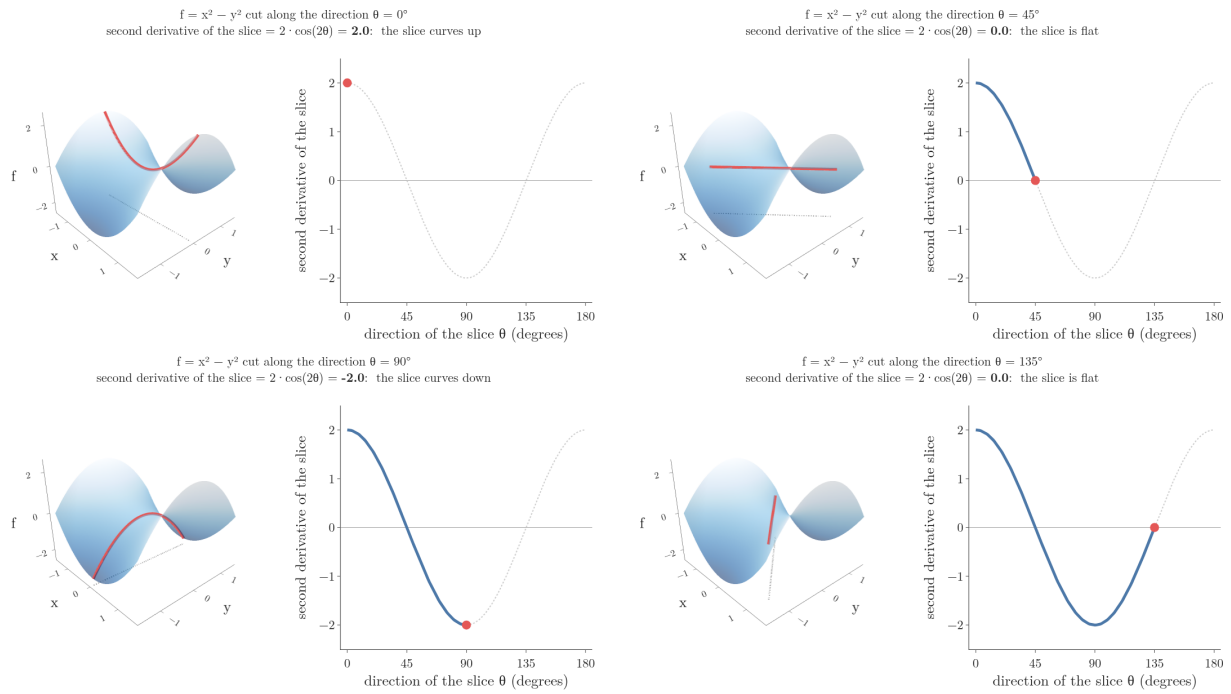


Figure 6: The saddle $f = x^2 - y^2$ cut by a vertical plane that turns from the x direction (angle 0) to the y direction (90 degrees). Left: the slice (red) on the surface. Right: the second derivative of the slice, $2\cos 2\theta$, traced against the angle: positive, zero, then negative. Idea after Khan Academy, "Saddle points"

The two axis slices are not enough. Take $f = x^2 + y^2 + p\,xy$ with a number p we can turn up. Its **second partial derivatives** (G-1760) along the axes are $f_{xx} = 2$ and $f_{yy} = 2$ for every p: both axis slices always bend up. Yet Figure 7 shows the bowl turning into a saddle as p grows.

1. Walk a distance s along the diagonal direction $(1, -1)/\sqrt{2}$. The point reached has

$$x = s/\sqrt{2}$$

$$y = -s/\sqrt{2}$$

   One term per line:

$$x^2 + y^2 = \tfrac{s^2}{2} + \tfrac{s^2}{2} = s^2$$

$$p\,xy = p \times \tfrac{s}{\sqrt{2}} \times \left(-\tfrac{s}{\sqrt{2}}\right) = -\tfrac{p}{2}\,s^2$$

$$f = \left(1 - \tfrac{p}{2}\right) s^2$$

   Its second derivative with respect to s is

$$2\left(1 - \tfrac{p}{2}\right) = 2 - p$$

2. For $p < 2$ it still bends up: a bowl.
3. At $p = 2$ it is flat.

4. For $p > 2$ it bends down while the axis slices bend up: a saddle.

The term that does this is the mixed one: $f_{xy} = p$. The **mixed partial derivative** (G-1236) measures how much the two diagonal directions disagree. The purest case is $f = xy$: flat along both axes ($f_{xx} = f_{yy} = 0$), bending up along one diagonal and down along the other.

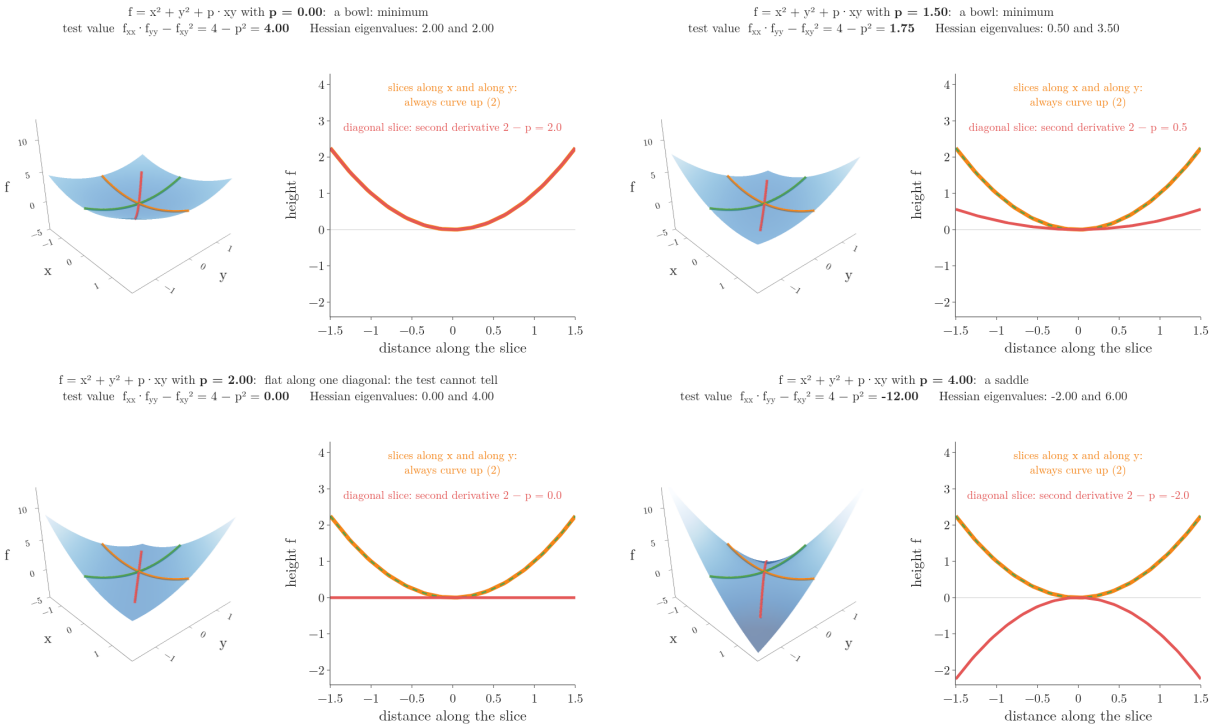


Figure 7: The surface $f = x^2 + y^2 + p\,xy$ as p grows from 0 to 4. The slices along x and y (orange, green) never change; the diagonal slice (red) flattens at $p = 2$ and then bends down, so the bowl becomes a saddle. The title tracks the test value $4 - p^2$ and the Hessian's eigenvalues $2 - p$ and $2 + p$. Idea after Khan Academy, "Second partial derivative test"

The second partial derivative test (G-2253). One number weighs the axis slices against the mixed term.

1. **In words:** multiply the two axis bends and subtract the square of the mixed partial derivative.
2. **Formula:** at a stationary point, compute

$$D = f_{xx}\,f_{yy} - f_{xy}^2$$

   - $D > 0$ and $f_{xx} > 0$: a minimum;
   - $D > 0$ and $f_{xx} < 0$: a maximum;
   - $D < 0$: a saddle point;
   - $D = 0$: the test gives no answer.
3. **Example:** for $f = x^2 + y^2 + p\,xy$ at the origin, $f_{xx} = 2$, $f_{yy} = 2$ and $f_{xy} = p$:

$$D = 2 \times 2 - p^2 = 4 - p^2$$

With $p = 0$, $D > 0$ and $f_{xx} > 0$, so a minimum:

$$D = 4 - 0 = 4$$

With $p = 4$, $D < 0$, so a saddle:

$$D = 4 - 16 = -12$$

With $p = 2$, $D = 0$, the switch of Figure 7:

$$D = 4 - 4 = 0$$

The running example $f = x^3 + xy + y^2$ has two stationary points. We find them by setting both partial derivatives to zero and solving, one line at a time.

$$\frac{\partial f}{\partial x} = 3x^2 + y = 0 \qquad \text{(equation 1)}$$

$$\frac{\partial f}{\partial y} = x + 2y = 0 \qquad \text{(equation 2)}$$

Equation 2 gives x in terms of y:

$$x = -2y$$

Put this into equation 1:

$$3(-2y)^2 + y = 0$$

$$12y^2 + y = 0$$

$$y\,(12y + 1) = 0$$

A product is zero when one of its factors is zero, so

$$y = 0 \quad \text{or} \quad y = -\tfrac{1}{12}$$

and from $x = -2y$:

$$y = 0 : \quad x = 0$$

$$y = -\tfrac{1}{12} : \quad x = -2 \times \left(-\tfrac{1}{12}\right) = \tfrac{1}{6}$$

The two stationary points are $(0,0)$ and $(\frac{1}{6}, -\frac{1}{12})$. Now the test at each. The Hessian is $H = \begin{bmatrix} 6x & 1 \\ 1 & 2 \end{bmatrix}$ (Section 5.1).

At $(0,0)$:

$$f_{xx} = 6 \times 0 = 0$$

$$f_{yy} = 2$$

$$f_{xy} = 1$$

$$D = 0 \times 2 - 1^2 = -1 < 0$$

a saddle point.

At $(\frac{1}{6}, -\frac{1}{12})$:

$$f_{xx} = 6 \times \tfrac{1}{6} = 1$$

$$f_{yy} = 2$$

$$f_{xy} = 1$$

$$D = 1 \times 2 - 1^2 = 1 > 0$$

and $f_{xx} = 1 > 0$: a minimum.

The same test with eigenvalues. D is the determinant of the Hessian. For a 2×2 matrix the determinant is the product of the two eigenvalues, so $D > 0$ says the two eigenvalues have the same sign and $D < 0$ says their signs differ. The eigenvalue form works for any number of inputs, where a single number D is no longer enough. The Hessian's eigenvectors are the slice directions of strongest and weakest bending, and its eigenvalues are the second derivatives of those two slices (see finding the eigenvectors). For $x^2 + y^2 + p\,xy$ the Hessian has rows $[2, p]$ and $[p, 2]$; its eigenvectors are the two diagonals and its eigenvalues are $2 + p$ and $2 - p$, the red slice of Figure 7. Figure 8 shows the three basic shapes.

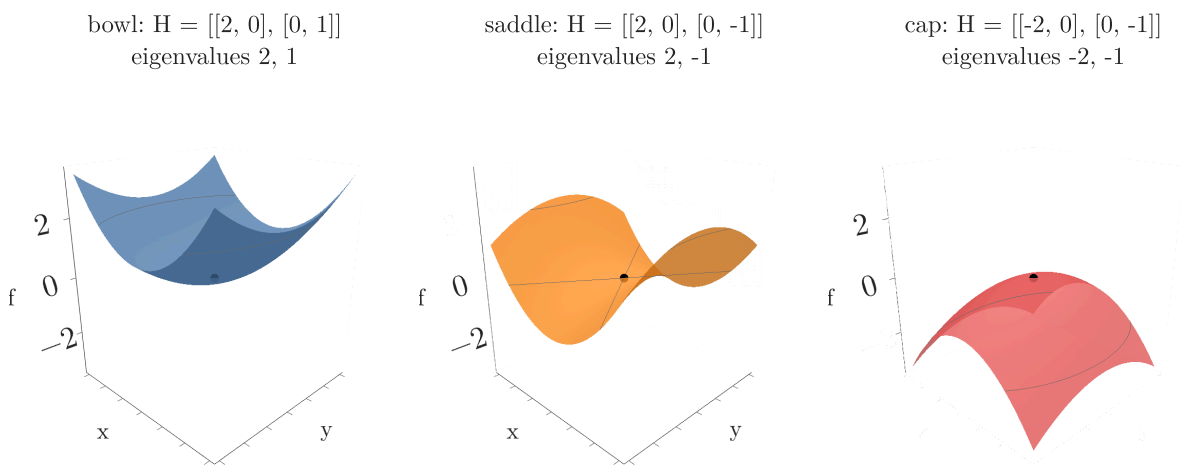


Figure 8: Three surfaces and the eigenvalues of their Hessians: both positive (bowl), opposite signs (saddle), both negative (cap)

- **Bowl:** all eigenvalues positive. The surface curves up in every direction; a point with zero gradient is a minimum.
- **Saddle** (a saddle point when the gradient is zero): positive and negative eigenvalues. The surface curves up in some directions and down in others; a point with zero gradient is neither a minimum nor a maximum.
- **Cap:** all eigenvalues negative. The surface curves down everywhere; a point with zero gradient is a maximum.

For the bowl of Section 5.2, H has rows $[2, 1]$ and $[1, 4]$. Its eigenvalues λ are the numbers that make $\det(H - \lambda I) = 0$ (see finding eigenvalues), one step per line:

$$(2 - \lambda)(4 - \lambda) - 1 \times 1 = 0$$

$$\lambda^2 - 6\lambda + 8 - 1 = 0$$

$$\lambda^2 - 6\lambda + 7 = 0$$

$$\lambda = \frac{6 \pm \sqrt{36 - 28}}{2}$$

$$= \frac{6 \pm \sqrt{8}}{2} = 3 \pm \sqrt{2}$$

$$\lambda_1 = 3 + 1.41 = 4.41$$

$$\lambda_2 = 3 - 1.41 = 1.59$$

Both are positive, so the function is a bowl, as its contour map showed. The ratio

$$\frac{4.41}{1.59} = 2.8$$

says the bowl curves almost three times as steeply in one direction as in the other: its contour ellipses are stretched. The narrow valleys that slow down gradient descent (the effect of feature scaling) are surfaces whose Hessian has a large eigenvalue ratio.

> **Extra:** A function whose Hessian has no negative eigenvalues at any point is convex (Boyd and Vandenberghe §3.1.4): a single bowl with no local minima to get stuck in, the property the shape of the loss function asked of a loss. The least-squares Hessian $2\Phi^\mathsf{T}\Phi$ never has negative eigenvalues, because $\boldsymbol{\delta}^\mathsf{T}\Phi^\mathsf{T}\Phi\boldsymbol{\delta} = \|\Phi\boldsymbol{\delta}\|^2 \geq 0$ for every $\boldsymbol{\delta}$. This Hessian is why linear regression's loss is convex.

6. The multivariate Taylor series

> **Key point:** Keep adding terms: each new term uses the next higher derivatives at the point and one more copy of the step. The second-order term is half of (step times Hessian times step).

6.1 The terms up to second order

> **Key point:** Value, plus gradient times step, plus half the step times Hessian times step.

The tangent plane is the start of a series, exactly as in one variable, and the **quadratic approximation** (G-2252) of Section 4 is its next step. Section 4.2 found the bending terms one constant at a time; with the Hessian they are one line.

The step is a column, the difference of the two points:

$$\boldsymbol{\delta} = \mathbf{x} - \mathbf{x}_0$$

In our example $\boldsymbol{\delta} = [0.1, -0.1]^\mathsf{T}$, from $(1, 1)$ to $(1.1, 0.9)$. Its transpose $\boldsymbol{\delta}^\mathsf{T}$ is the same two numbers as a row, $[0.1, -0.1]$. The number of inputs is n, as everywhere in these Notes; here $n = 2$. The shapes column says how many rows and columns each factor has.

Order k	Term	Shapes	One-variable version
0	$f(\mathbf{x}_0)$	number	$f(x_0)$
1	$\nabla f(\mathbf{x}_0)\,\boldsymbol{\delta}$	$(1 \times n)(n \times 1)$	$f'(x_0)\,\delta$
2	$\frac{1}{2}\boldsymbol{\delta}^\mathsf{T} H(\mathbf{x}_0)\,\boldsymbol{\delta}$	$(1 \times n)(n \times n)(n \times 1)$	$\frac{1}{2}f''(x_0)\,\delta^2$

1. **In words:** the second-order term weighs every pair of step components by the matching Hessian entry, adds them up and halves the sum. With two inputs there are four pairs, one per Hessian entry:

Hessian entry	Pair of step components	Product
row 1, column 1: 6	$\delta_1\delta_1 = 0.1 \times 0.1 = 0.01$	$6 \times 0.01 = 0.06$
row 1, column 2: 1	$\delta_1\delta_2 = 0.1\times(-0.1) = -0.01$	$1 \times (-0.01) = -0.01$
row 2, column 1: 1	$\delta_2\delta_1 = -0.01$	$1 \times (-0.01) = -0.01$
row 2, column 2: 2	$\delta_2\delta_2 = (-0.1)^2 = 0.01$	$2 \times 0.01 = 0.02$

   The four products add to

$$0.06 - 0.01 - 0.01 + 0.02 = 0.06$$

   The double sum $\sum_i \sum_j$ below means "add over every row i and every column j", that is, over the four rows of this table.

2. **Formula:** the second-order **multivariate Taylor polynomial** (G-1284) is

$$T_2(\mathbf{x}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)\,\boldsymbol{\delta}$$

$$+\frac{1}{2}\,\boldsymbol{\delta}^{\mathsf{T}} H(\mathbf{x}_0)\,\boldsymbol{\delta}$$

   where

$$\boldsymbol{\delta}^{\mathsf{T}} H \boldsymbol{\delta} = \sum_i \sum_j H_{ij}\,\delta_i\,\delta_j$$

3. **Example:** at $(1, 1)$ with $\boldsymbol{\delta} = (0.1, -0.1)$ and H with rows $[6, 1]$ and $[1, 2]$, the table above gives

$$\boldsymbol{\delta}^{\mathsf{T}} H \boldsymbol{\delta} = 0.06 - 0.01 - 0.01 + 0.02$$

$$= 0.06$$

$$\tfrac{1}{2} \times 0.06 = 0.03$$

$$T_2 = 3.1 + 0.03 = 3.13$$

   against the true 3.131.

Figure 1 (right) shows T_2: a curved surface that follows f far beyond the region where the tangent plane was good.

Python: The three terms in NumPy.

```
import numpy as np

f0 = 3.0
g = np.array([4.0, 3.0])                  # gradient at (1, 1)
H = np.array([[6.0, 1.0], [1.0, 2.0]])    # Hessian at (1, 1)
```

```
d = np.array([0.1, -0.1])                    # step to (1.1, 0.9)

f0 + g @ d                          # 3.1   (first order)
f0 + g @ d + 0.5 * d @ H @ d        # 3.13  (second order)
```

6.2 Higher orders: tensors and outer products

> **Key point:** The k-th term multiplies every k-th derivative by the matching product of k step components, adds them all, and divides by $k!$. For $k = 2$ this is the Hessian table of Section 6.1.

The pattern continues. Section 6.1 needed one Hessian entry for every pair of inputs, such as (row 1, column 2). The third-order term needs one third derivative for every triple of inputs, such as (1, 1, 2): differentiate by x, by x, then by y. In general, the k-th order term needs one k-th derivative for every list of k input numbers $(i_1, i_2, \ldots, i_k)$, each between 1 and n.

An instance, $k = 3$, for our running example $f = x^3 + xy + y^2$ at $(1, 1)$ with $\boldsymbol{\delta} = (0.1, -0.1)$. With $n = 2$ inputs there are 8 triples (2 choices for each of the 3 places). Every third derivative is zero except one:

$$\frac{\partial^3 f}{\partial x^3} = \frac{\partial}{\partial x}\frac{\partial}{\partial x}(3x^2 + y)$$

$$= \frac{\partial}{\partial x}\, 6x = 6$$

Its triple is $(1, 1, 1)$, and the matching product of step components is

$$\delta_1\delta_1\delta_1 = 0.1^3 = 0.001$$

So the sum over all eight triples has one non-zero term, and the third-order term is

$$\frac{6 \times 0.001}{3!} = \frac{0.006}{6} = 0.001$$

The general form. The k-th derivatives at $\mathbf{x}_0$, all together, are written $D^k f(\mathbf{x}_0)$ (the letter D here stands for "derivative"; it is not the test number D of Section 5.3): an array with one entry per list of k indices, $n \times n \times \cdots \times n$ (k times). An array with more than two indices is a tensor. The array is combined with k copies of $\boldsymbol{\delta}$.

1. **In words:** multiply each k-th derivative entry by the matching product of step components, add them all, and divide by $k!$.
2. **Formula:** the **multivariate Taylor series** (G-1285) is

$$f(\mathbf{x}) = \sum_{k=0}^{\infty} \frac{D^k f(\mathbf{x}_0)\, \boldsymbol{\delta}^k}{k!}$$

where the numerator is the sum over every list of k indices:

$$D^k f(\mathbf{x}_0)\,\boldsymbol{\delta}^k =$$

$$\sum_{i_1=1}^{n} \cdots \sum_{i_k=1}^{n} D^k f(\mathbf{x}_0)[i_1, \ldots, i_k]\ \delta_{i_1} \cdots \delta_{i_k}$$

Check with $k = 3$ above: the only non-zero entry is $[1, 1, 1] = 6$, which gives

$$\frac{6 \times 0.001}{6} = 0.001$$

Stopping after the $k = m$ term gives the Taylor polynomial T_m of order m (here m counts orders; n still counts inputs).

3. **Example:** for $k = 2$, $\boldsymbol{\delta}^2$ means the **outer product** (G-1418) $\boldsymbol{\delta}\boldsymbol{\delta}^\mathsf{T}$, the matrix with entries $\delta_i \delta_j$. With $\boldsymbol{\delta} = (0.1, -0.1)$:

$$\boldsymbol{\delta}\boldsymbol{\delta}^\mathsf{T} = \begin{bmatrix} 0.01 & -0.01 \\ -0.01 & 0.01 \end{bmatrix}$$

Multiplying entry by entry with H, one product per line:

$$6 \times 0.01 = 0.06$$

$$1 \times (-0.01) = -0.01$$

$$1 \times (-0.01) = -0.01$$

$$2 \times 0.01 = 0.02$$

$$0.06 - 0.01 - 0.01 + 0.02 = 0.06$$

the same $\boldsymbol{\delta}^\mathsf{T} H \boldsymbol{\delta}$ as before.

Each extra copy of $\boldsymbol{\delta}$ adds one index: $\boldsymbol{\delta}\boldsymbol{\delta}^\mathsf{T}$ is a matrix (2 indices), and three copies give a 3-index tensor. The symbol $\otimes$ (outer product) writes it as $\boldsymbol{\delta} \otimes \boldsymbol{\delta} \otimes \boldsymbol{\delta}$; its entries are all products $\delta_i \delta_j \delta_k$. For our step, the $(1, 1, 1)$ entry is $0.1^3 = 0.001$, the number used in the $k = 3$ instance above, and the $(1, 1, 2)$ entry is the product below (see tensors from 0D to 5D):

$$0.1 \times 0.1 \times (-0.1) = -0.001$$

6.3 A complete example: the expansion is exact for a polynomial

> **Key point:** A polynomial of degree 3 needs exactly the terms up to third order; then the Taylor polynomial is the function itself.

As in one variable (a polynomial is its own Taylor polynomial), the Taylor series of a polynomial stops. For $f(x, y) = x^3 + xy + y^2$ at $(1, 1)$, write $\delta_x = x - 1$ and $\delta_y = y - 1$.

- **Order 0:** $f(1, 1) = 3$.
- **Order 1:** $\nabla f(1, 1) = [4, 3]$ gives $4\delta_x + 3\delta_y$.

- **Order 2:** the term from Section 5.1, $3\delta_x^2 + \delta_x\delta_y + \delta_y^2$.
- **Order 3:** of all third partial derivatives only $\partial^3 f/\partial x^3 = 6$ is not zero, so the term is δ_x^3 (6 divided by $3! = 6$).
- **Order 4 and up:** all derivatives are zero.

1. **In words:** add the four orders.
2. **Formula:**
$$f(x, y) = 3 + 4\delta_x + 3\delta_y + 3\delta_x^2 + \delta_x\delta_y + \delta_y^2 + \delta_x^3$$
3. **Example:** at $(1.1, 0.9)$, so $\delta_x = 0.1$ and $\delta_y = -0.1$, the running totals are:
$$T_1 = 3 + 0.4 - 0.3 = 3.1$$
$$T_2 = 3.1 + 0.03 = 3.13$$
$$T_3 = 3.13 + 0.1^3 = 3.13 + 0.001 = 3.131$$
The function itself, one term per line:
$$x^3 = 1.1^3 = 1.331$$
$$xy = 1.1 \times 0.9 = 0.99$$
$$y^2 = 0.9^2 = 0.81$$
$$f(1.1, 0.9) = 1.331 + 0.99 + 0.81 = 3.131$$
Exactly T_3. Further away, at $(1.5, 0.5)$, so $\delta_x = 0.5$ and $\delta_y = -0.5$:
$$T_1 = 3 + 4 \times 0.5 + 3 \times (-0.5) = 3.5$$
The second-order terms:
$$3 \times 0.25 + 0.5 \times (-0.5) + 0.25 = 0.75$$
$$T_2 = 3.5 + 0.75 = 4.25$$
$$T_3 = 4.25 + 0.5^3 = 4.25 + 0.125 = 4.375$$
$$f(1.5, 0.5) = 3.375 + 0.75 + 0.25 = 4.375$$
The low orders are worse there, but the full expansion is still exact.

Multiplying out the brackets gives back $x^3 + xy + y^2$; we checked this symbolically.

Figure 10 below walks a straight path from $(1, 1)$ to $(1.5, 0.5)$. First, the landscape that path crosses. Figure 9 draws $f(x, y) = x^3 + xy + y^2$ near $(1, 1)$, where $f(1, 1) = 3$, as a surface (left): a sheet that rises towards large x and y, and rises faster along x because of the x^3 term. The path from $(1, 1)$ to $(1.5, 0.5)$ climbs it to $f(1.5, 0.5) = 4.375$. On the right is the same sheet seen from straight above, the contour map used in Figure 10:

- each line joins points of the same height; darker = lower;
- lines close together = steep: they crowd at the top right, where the sheet climbs fastest.

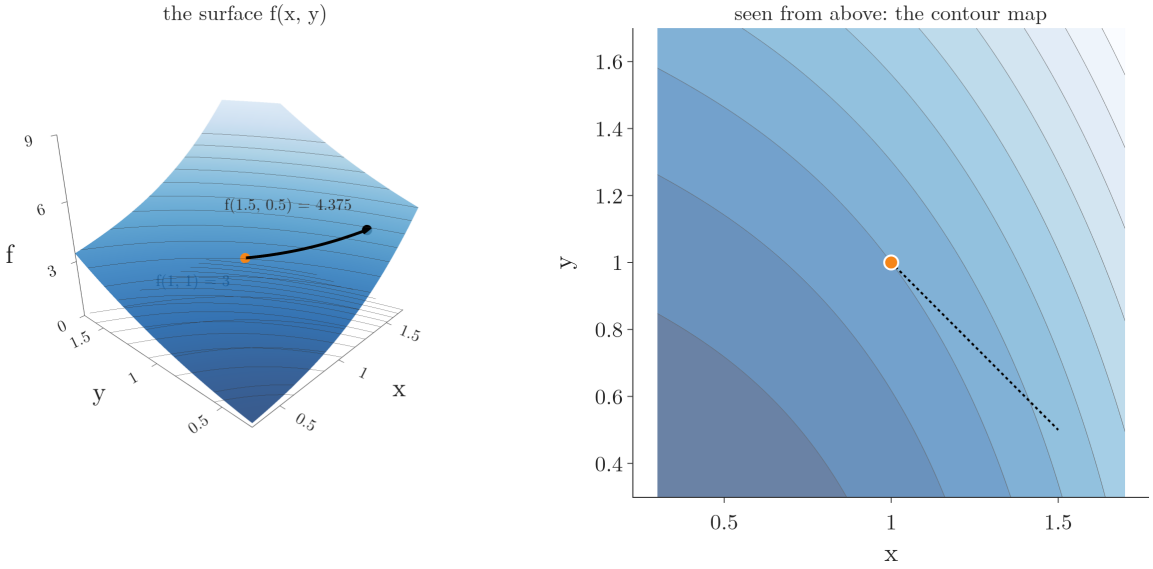


Figure 9: The surface $f(x, y) = x^3 + xy + y^2$ near $(1, 1)$ (left), with its contour lines drawn on it and dropped to the floor, and the same surface seen from above as a contour map (right). Same colours, same square, same path from $(1, 1)$ (orange) to $(1.5, 0.5)$

Figure 10 walks along the straight path from $(1, 1)$ to $(1.5, 0.5)$ and reads all three polynomials on the way. Near the start they agree with f. Further out the tangent plane T_1 falls behind first, then T_2; T_3 stays on f the whole way.

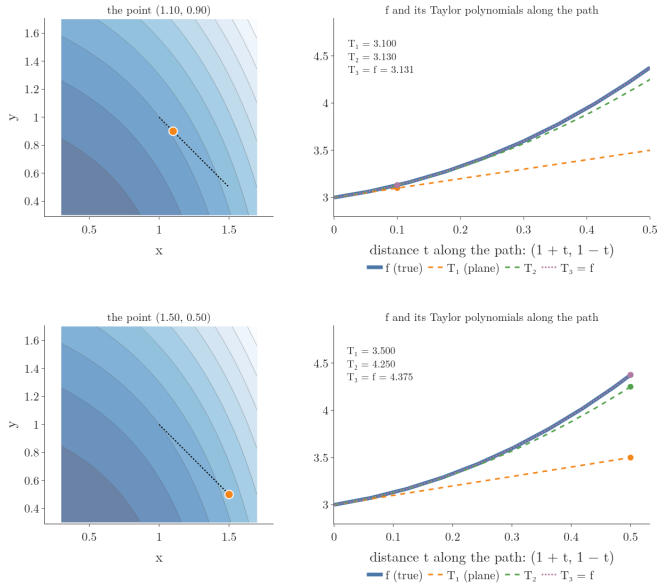


Figure 10: The Taylor polynomials of $f(x, y) = x^3 + xy + y^2$ around $(1, 1)$ along the path to $(1.5, 0.5)$. Left: the moving point on the contour map of Figure 9. Right: f (thick blue) against T_1, T_2 and T_3; T_3 equals f.

7. Where ML uses second-order approximations

Key point: Minimising the second-order Taylor polynomial instead of the function itself is **Newton's method** (G-1321); XGBoost does this for every tree, and other methods use it to approximate distributions.

The second-order polynomial is a bowl (when H has positive eigenvalues), and the bottom of a bowl has a formula. The formula makes T_2 a useful stand-in for a function that is hard to minimise directly.

Extra: Newton's method in several variables. Setting the gradient of T_2 with respect to $\boldsymbol{\delta}$ to zero gives $\nabla f^{\mathsf{T}} + H\boldsymbol{\delta} = \mathbf{0}$, so the step to the bottom of the local bowl is

$$\boldsymbol{\delta} = -H^{-1}\nabla f^{\mathsf{T}}$$

For the bowl of Section 5.2 at $(1, 1)$, $\nabla f = [3, 5]$ and H has rows $[2, 1]$ and $[1, 4]$. Solve $H\boldsymbol{\delta} = -[3, 5]^{\mathsf{T}}$ for $\boldsymbol{\delta} = (a, b)$, one line at a time:

$$2a + b = -3 \qquad \text{(row 1)}$$

$$a + 4b = -5 \qquad \text{(row 2)}$$

$$a = -5 - 4b \qquad \text{(from row 2)}$$

$$2(-5 - 4b) + b = -3$$

$$-10 - 7b = -3$$

$$b = -1$$

$$a = -5 - 4 \times (-1) = -1$$

So $\boldsymbol{\delta} = (-1, -1)$, landing exactly on the minimum $(0, 0)$ in one step: for a quadratic function T_2 is the function itself. Gradient descent, which only uses the tangent plane, needs many small steps for the same trip. With one variable this is the **Newton step** of the leaf values in log-odds, and XGBoost's leaf formula $-G/(H + \lambda)$ in the best leaf output is a Newton step per leaf (Chen and Guestrin §2.2). The catch: with millions of parameters the Hessian has millions squared entries, which is why deep learning mostly stays with first-order methods (Goodfellow et al. §8.6.1).

Extra: Quasi-Newton methods (G-1603): **BFGS** (G-283) and **L-BFGS** (G-1023). These skip the Hessian and build a stand-in matrix B from gradients alone. After each step, the step $\mathbf{s}$ and the gradient change $\mathbf{y}$ are measured:

$$\mathbf{s} = \mathbf{x}_{k+1} - \mathbf{x}_k$$

$$\mathbf{y} = \nabla f_{k+1} - \nabla f_k$$

B is updated so that

$$B_{k+1}\,\mathbf{s} = \mathbf{y}$$

The condition is the **secant equation** (G-1757): the new B must reproduce the gradient change just seen. With one variable it says B is the slope between two gradient readings:

$$B = \frac{f'(x_{k+1}) - f'(x_k)}{x_{k+1} - x_k}$$

For $f = x^3$ stepping from $x = 1$ to $x = 2$, f' goes from 3 to 12:

$$B = \frac{12 - 3}{2 - 1} = 9$$

This lies between the true **curvatures** (G-521) $f''(1) = 6$ and $f''(2) = 12$. The step is then $\boldsymbol{\delta} = -B^{-1}\nabla f^{\mathsf{T}}$, as in Newton's method, usually shortened by a line search (trying a shorter step until the function drops enough). **BFGS** (Broyden, Fletcher, Goldfarb, Shanno) is the most used update rule; it keeps B symmetric and positive definite (all eigenvalues positive), so every step goes downhill. **L-BFGS** ("limited memory") stores only the last few $(\mathbf{s}, \mathbf{y})$ pairs instead of the full $n \times n$ matrix, which makes it usable with many parameters. L-BFGS is the default solver of `LogisticRegression` (the solver). (Nocedal and Wright, ch. 6 for BFGS, §7.2 for L-BFGS.)

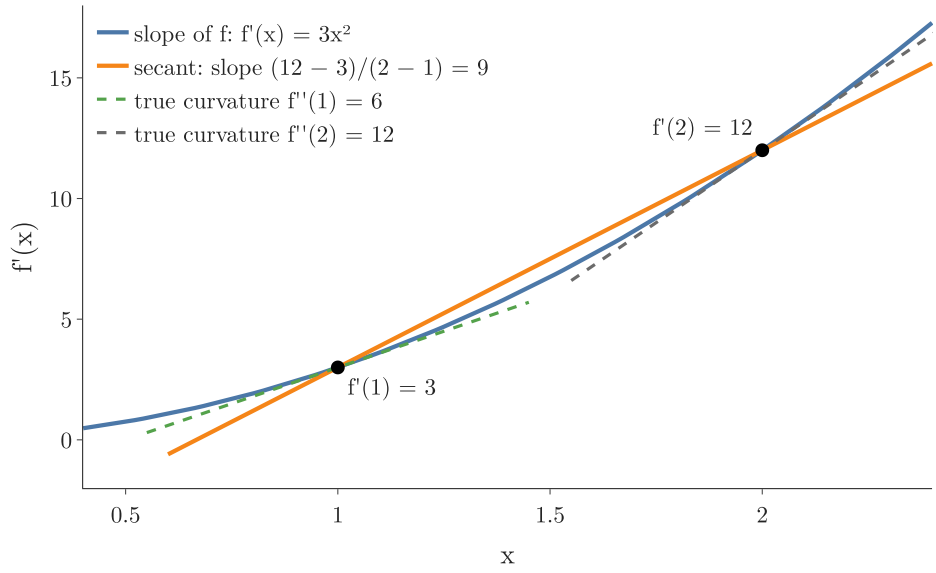


Figure 11: The secant idea for $f = x^3$: the line through the slope readings at $x = 1$ and $x = 2$ has slope 9, a stand-in for the true curvatures 6 and 12

Figure 11 shows the one-variable case: BFGS never computes the curvature (dashed), it reads the slope twice and takes the line through the two readings (orange).

The race below runs on a new function, the curved valley. Its formula is

$$f(x, y) = (1 - x)^2 + 5(y - x^2)^2$$

At the starting point $(-1.2, 1)$, one term per line:

$$(1 - (-1.2))^2 = 2.2^2 = 4.84$$

$$5 \times (1 - 1.44)^2 = 5 \times 0.1936 = 0.968$$

$$f(-1.2, 1) = 4.84 + 0.968 = 5.808$$

At $(1, 1)$ both terms are 0, so $f(1, 1) = 0$, the lowest value. Figure 12 shows its shape (left). The second term is 0 along the curve $y = x^2$ (dashed), so the surface has a long, narrow valley floor that bends along that curve, with steep walls on both sides. Along the floor, only the small first term is left, so the floor slopes gently down to the minimum at $(1, 1)$. On the right is the same surface seen from above, the contour map of the race (Figure 13): darker = lower, and the lines crowd together on the steep walls.

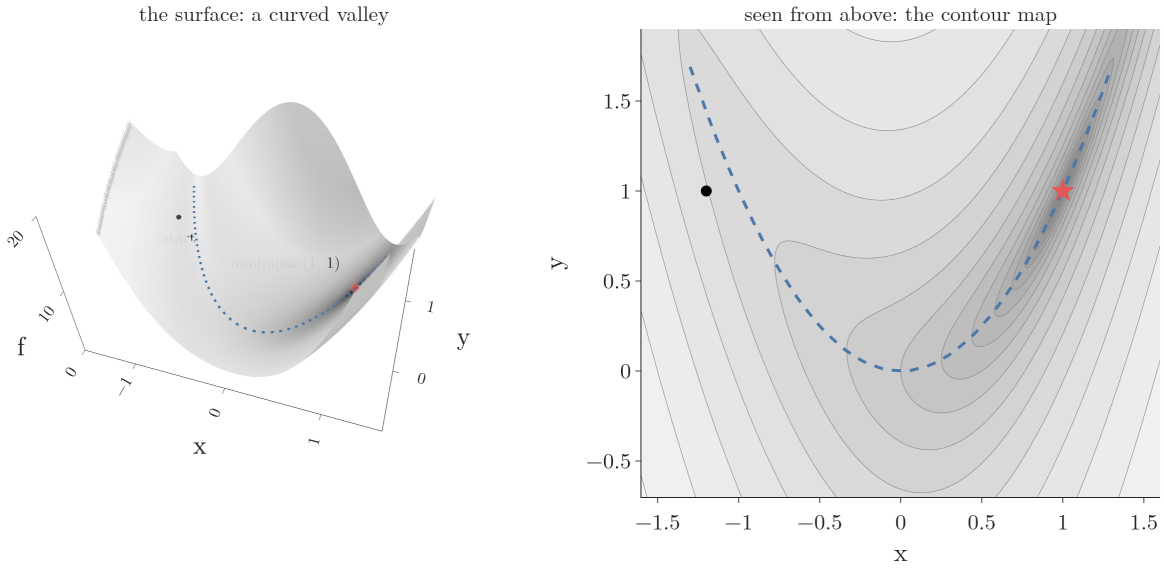


Figure 12: The curved valley $f(x, y) = (1 - x)^2 + 5(y - x^2)^2$ as a surface (left) and seen from above as a contour map (right). Same colours (darker = lower), same square. Black dot: the start $(-1.2, 1)$; red: the minimum $(1, 1)$; dashed: the valley floor $y = x^2$

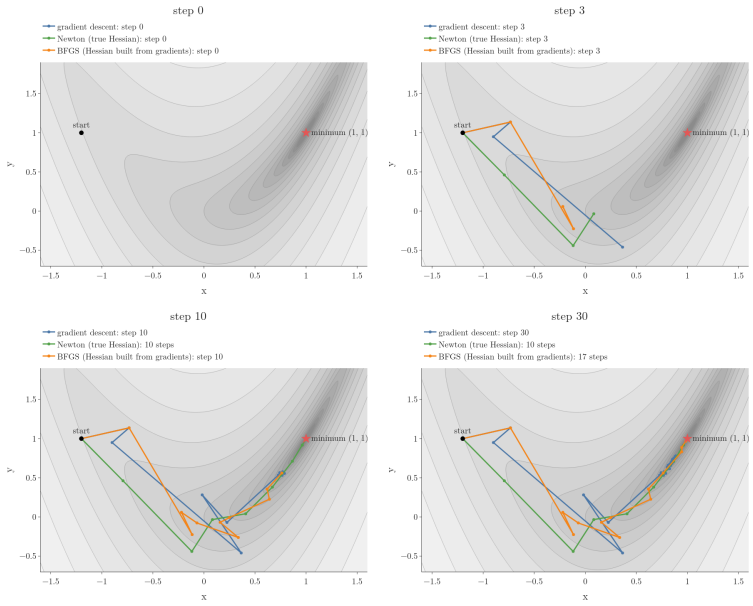


Figure 13: Gradient descent, Newton and BFGS walking down the same curved valley from $(-1.2, 1)$ to the minimum $(1, 1)$, one step per frame

Figure 13 races the three methods on the curved valley above, all with the same rule for the step length. Newton, with the true Hessian, needs 10 steps. BFGS, which only ever sees gradients, needs 17: its first steps wander while B is still a guess, then it settles into the valley like Newton. The figure script checks this: for steps 0 to 6, B is still 90% or more away from the true Hessian, and from step 7 on it is within about 30%. Gradient descent zig-zags across the narrow valley and needs 717 steps, because the gradient crosses the contour lines at right angles (the gradient as an arrow): on the steep walls it points mostly across the valley, not along the gently sloping floor towards $(1, 1)$.

> **Extra:** Second-order Taylor expansions also approximate probability distributions. The **Laplace approximation** (G-1043) replaces a distribution near its peak by a normal distribution whose spread comes from the Hessian of its log there (Bishop §4.4). The **extended Kalman filter**, used to track moving objects, linearises a nonlinear system at every time step with the first-order expansion (Thrun et al. §3.3).

8. Summary

Object	Formula	Example: $f = x^3 + xy + y^2$ at $(1, 1)$	Why it matters
Mixed partial derivatives	$\partial^2 f/\partial x \partial y = \partial^2 f/\partial y \partial x$	both equal 1	makes the Hessian symmetric
Quadratic approximation	tangent plane $+ \frac{1}{2} f_{xx}\delta_x^2 + f_{xy}\delta_x\delta_y + \frac{1}{2} f_{yy}\delta_y^2$	3.13 at $(1.1, 0.9)$	bends with f, so it stays close over a wider area than the plane
Hessian	$H_{ij} = \partial^2 f/\partial x_i \partial x_j$, symmetric $n \times n$	rows $[6, 1]$, $[1, 2]$	describes how the surface curves near the point
Second partial derivative test	$D = f_{xx}f_{yy} - f_{xy}^2$ at a stationary point	$(0, 0)$: $D = -1$, saddle; $(\frac{1}{6}, -\frac{1}{12})$: $D = 1$, minimum	tells a minimum from a saddle or a maximum where the gradient is zero
Tangent plane (first order)	$f(\mathbf{x}_0) + \nabla f\, \boldsymbol{\delta}$	3.1 at $(1.1, 0.9)$	uses only the gradient; good only near the point
Second-order Taylor	$+ \frac{1}{2}\boldsymbol{\delta}^{\mathsf{T}} H \boldsymbol{\delta}$	3.13	a bowl whose bottom has a formula
Third-order Taylor	$+ \frac{1}{3!} D^3 f\, \boldsymbol{\delta}^3$	3.131 (exact)	exact for a cubic, whose higher derivatives are zero
Newton step	$\boldsymbol{\delta} = -H^{-1}\nabla f^{\mathsf{T}}$	bowl: $(1, 1) \to (0, 0)$	reaches the bottom of a quadratic in one step, where gradient descent needs many
Secant equation (BFGS)	$B_{k+1}\mathbf{s} = \mathbf{y}$	x^3, $1 \to 2$: $B = 9$	a curvature stand-in built from gradients alone, with no Hessian

- **Second partial derivatives** (G-1760) can be taken in either order; the Hessian is therefore symmetric.
- The Hessian measures curvature; the signs of its eigenvalues separate bowls, saddles and caps, because a zero gradient alone cannot tell a minimum from a hilltop or a saddle. For two inputs **the second partial derivative test** (G-2253) reads the same from $f_{xx}f_{yy} - f_{xy}^2$.
- The **mixed partial derivative** (G-1236) can turn a bowl into a saddle even when both axis slices bend up, so checking the two axis slices is not enough.
- Linearisation replaces a surface by its tangent plane, good only near the point, because

a plane cannot bend.

- The **multivariate Taylor series** (G-1285) adds terms with k copies of the step and the k-th derivative tensor; the second-order term is $\frac{1}{2}\boldsymbol{\delta}^{\mathsf{T}} H \boldsymbol{\delta}$, so the second-order polynomial is a bowl that **Newton's method** (G-1321), and XGBoost for every tree, can minimise directly.
- A polynomial of degree k is reproduced exactly by its Taylor polynomial of degree k, because all its derivatives above order k are zero.
- So the gradient gives the flat approximation of a surface near a point, and the Hessian bends it to follow the surface's **curvature** (G-521).

9. Sources

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10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Multivariate Taylor polynomial (G-1284)	An approximation of a function of several variables near a point, built from its derivatives there up to order n (the multivariate Taylor series cut after the $k = n$ term); the second-order one uses the gradient and the Hessian.
Tangent plane (G-1946)	The flat plane that touches a surface at one point with the same slopes (gradient) there; it approximates the surface near that point and is the first-order Taylor approximation.
Second partial derivative (G-1760)	A partial derivative taken of a partial derivative, such as $\partial^2 f/\partial x^2$; it measures how the slope along one input changes, that is, how the surface bends.
Quadratic approximation (G-2252)	A curved surface that matches a function near a point in value, slopes and bending: the tangent plane plus the terms $\frac{1}{2}f_{xx}\delta_x^2 + f_{xy}\delta_x\delta_y + \frac{1}{2}f_{yy}\delta_y^2$; it is the second-order Taylor polynomial, and Newton's method steps to its minimum.
Mixed partial derivative (G-1236)	A second partial derivative taken with respect to two different variables, such as $\partial^2 f/\partial y\,\partial x$: how the slope along x changes as y changes; the off-diagonal entries of the Hessian.
Curvature (G-521)	How fast the slope of a surface changes; given in each direction by the Hessian.
Hessian matrix (G-888)	The table of how a many-input function's slopes change, which measures its curvature: the symmetric $n \times n$ matrix of all second partial derivatives of $f : \mathbb{R}^n \to \mathbb{R}$.

Term	Meaning
Second partial derivative test (G-2253)	A test that tells what a stationary point of a function of two inputs is from one number, $D = f_{xx}f_{yy} - f_{xy}^2$: $D > 0$ a minimum or maximum (by the sign of f_{xx}), $D < 0$ a saddle point, $D = 0$ no answer.
Multivariate Taylor series (G-1285)	An approximation of a function of several variables near a point $\mathbf{x}_0$ from all its derivatives there, each term adding one more order of correction: $\sum_k D^k f(\mathbf{x}_0)\,\boldsymbol{\delta}^k/k!$.
Outer product (G-1418)	Multiplying a column vector by a row vector to get a whole matrix: $\boldsymbol{\delta}\boldsymbol{\delta}^\mathsf{T}$ has entry $\delta_i\delta_j$ for every pair; it is what $\boldsymbol{\delta}^2$ means in a Taylor polynomial of several variables.
Newton's method (G-1321)	A way to find a minimum: at the current point, fit a bowl (the second-order Taylor polynomial, from the first and second derivatives), jump to its lowest point, and repeat; the step is $\boldsymbol{\delta} = -H^{-1}\nabla f^\mathsf{T}$. XGBoost uses one such step for every tree.
Quasi-Newton method (G-1603)	Newton's method with the Hessian replaced by a stand-in matrix built from how the gradient changes between steps, so no second derivatives are computed; BFGS and L-BFGS are examples.
BFGS (G-283)	The most used way to take Newton-like steps without computing second derivatives (a quasi-Newton method): it builds a stand-in for the Hessian from gradients alone, kept symmetric and positive definite.
L-BFGS (G-1023)	An optimiser that uses the curvature of the loss without storing the full Hessian: it builds a stand-in for it from only the last few steps and gradient changes, saving memory (limited-memory BFGS, a quasi-Newton method). scikit-learn's default logistic regression solver.
Secant equation (G-1757)	The condition $B_{k+1}\mathbf{s} = \mathbf{y}$ on the matrix that stands in for the Hessian: the new matrix must reproduce the gradient change $\mathbf{y}$ seen over the last step $\mathbf{s}$, so it captures the curvature just observed.

Term	Meaning
Laplace approximation (G-1043)	Replacing a distribution near its peak with a normal distribution whose width comes from the curvature there (the Hessian).
Second derivative (G-2249)	The derivative of the derivative; it measures how the curve bends.
Observation (G-1374)	One record of a dataset: one row of the data table.
Contour plot (G-468)	A map of a surface seen from above, with lines joining points of equal height, so a 3D shape such as a loss bowl can be drawn and read on flat paper.
Stationary point (G-1879)	A point where a function is flat, its slope is zero (the derivative, or every partial derivative, is zero); it is a minimum, a maximum or a saddle point.
Saddle point (G-1718)	A flat point that curves up in one direction and down in another.